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# 晶圓廠 Surfscan 64×0 操作規範

## FAB Surfscan 64×0 Operation Instruction

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## Surfscan 64×0 操作規範

# Surfscan 64×0 Operation Instruction

### 壹、機台設備

#### I、Machine Equipment

一、機台設備為 TENCOR 製造，型號為 6400 與 6420 的 Surfscan。

A、Manufacture：TENCOR INSTRUMENTS. Model：Surfscan 6400 & 6420.

二、雷射光源為氬離子 ( Argon - ion) 雷射，波長為 4880A。

B、The light source is Argon-ion laser and the wave length is 4880A.

### 貳、適用時機及適用範圍

#### II、Time and Scope of Application

站別：(略)。

Operation code: (omitted)

### 參、製程目的

#### III、Purpose

測量 Poly - Si · TiN · Al · Polycide 等薄膜的 Particle。

Surfscan 64X0 is the machine for measuring particle count of thin film such as Poly-Si ,TiN,Al,Polycide.

### 肆、程式內容

|      |            |          |   |       |   |
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#### IV、Contents

一、程式名稱以薄膜名稱命名之，下列表格附有各程式名稱與其所適用之薄膜厚度。

A、There are several recipe names and the thickness for each recipe in the following sheet.

| Recipe ID  | Film on Top  | Thickness(A) | Note              |
|------------|--------------|--------------|-------------------|
| BARE_SI    | Si           | 0            |                   |
| TIN-36     | TiN          | 1.2K         |                   |
| TEOS-4600  | Therm Ox     | 4.6K         |                   |
| BPSG       | BPSG         | 10K          |                   |
| WSIX-1K    | Polycide     | 1K           |                   |
| DCS-1K     | Polycide-DCS | 1K           |                   |
| BARE-0.2UM | Si           | 0            | Pre-polycide Meas |
| QC_0.3     | Si           | 0            | In-come QC        |
| POLY-550   | POLY         | 550 ~ 1000   |                   |
| POLY-1700  | POLY         | 1001 ~ 3000  |                   |
| POLY-3800  | POLY         | 3001 ~ 5000  |                   |

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## 二、程式內容 Program Content。

| Recipe         | Gain | Edge | TOX  | 1     | 2     | 3     | 4     | 5     | 6     | 7    | 8    | MAX |
|----------------|------|------|------|-------|-------|-------|-------|-------|-------|------|------|-----|
| *IMP           | 6    | 4    | 400  | 0.2   | 0.282 | 0.397 | 0.559 | 0.787 | 1.11  | 1.56 | 2.20 | 3.1 |
| *1             | 5    | 4    |      | 0.3   | 0.402 | 0.538 | 0.72  | 0.964 | 1.29  | 1.73 | 2.31 | 3.1 |
| *3             | 6    | 4    | 400  | 0.3   | 0.402 | 0.538 | 0.72  | 0.964 | 1.29  | 1.73 | 2.31 | 3.1 |
| *5             | 5    | 4    | 1000 | 0.3   | 0.402 | 0.538 | 0.72  | 0.964 | 1.29  | 1.73 | 2.31 | 3.1 |
| *5-POLY        | 6    | 4    | 1000 | 0.215 | 0.389 | 0.56  | 0.807 | 1.16  | 1.67  | 2.41 | 2.74 | 3.1 |
| *7             | 5    | 4    | 2000 | 0.215 | 0.319 | 0.472 | 0.7   | 1.04  | 1.54  | 2.2  | 2.28 | 3.1 |
| *9             | 5    | 4    | 4500 | 0.23  | 0.318 | 0.441 | 0.610 | 0.844 | 1.17  | 1.62 | 2.24 | 3.1 |
| 18             | 5    | 4    |      | 1.14  | 1.29  | 1.46  | 1.66  | 1.88  | 2.13  | 2.41 | 2.74 | 3.1 |
| 19             | 5    | 4    | 1600 | 0.203 | 1.3   | 1.6   |       |       |       |      |      | 3.1 |
| 20             | 5    | 4    |      | 0.3   | 1.1   | 3     |       |       |       |      |      | 3.1 |
| *DIFF          | 6    | 4    | 400  | 0.50  | 0.628 | 0.798 | 0.991 | 1.24  | 1.56  | 1.96 | 2.47 | 3.1 |
| BARE-SI        | 5    | 4    |      | 0.3   | 0.402 | 0.538 | 0.72  | 0.964 | 1.29  | 1.73 | 2.31 | 3.1 |
| BARE-0.12UM    | 7    | 4    |      | 0.12  | 0.152 | 0.232 | 0.356 | 0.545 | 0.835 | 1.28 | 1.96 | 3.1 |
| BARE-0.2UM     | 5    | 4    |      | 0.2   | 0.282 | 0.397 | 0.559 | 0.787 | 1.11  | 1.56 | 2.2  | 3.1 |
| BPSG           | 5    | 4    | 5000 | 0.3   | 0.38  | 0.482 | 0.611 | 0.774 | 0.982 | 1.24 | 1.58 | 17  |
| CV-1000        | 5    | 4    | 1000 | 0.3   | 0.402 | 0.538 | 0.72  | 0.964 | 1.29  | 1.73 | 2.31 | 3.1 |
| OXIDE < 500    | 6    | 4    | 400  | 0.3   | 0.402 | 0.538 | 0.72  | 0.964 | 1.29  | 1.73 | 2.31 | 3.1 |
| POLY 1001-3000 | 5    | 4    | 1700 | 0.35  | 0.460 | 0.604 | 0.793 | 1.04  | 1.37  | 1.80 | 2.36 | 3.1 |
| POLY 3001-5000 | 5    | 4    | 3800 | 0.50  | 0.628 | 0.798 | 0.991 | 1.24  | 1.56  | 1.96 | 2.47 | 3.1 |
| POLY < 1000    | 5    | 4    | 550  | 0.35  | 0.460 | 0.604 | 0.793 | 1.04  | 1.37  | 1.80 | 2.36 | 3.1 |
| POLY > 5001    | 5    | 4    | 4200 | 0.9   | 1.05  | 1.23  | 1.43  | 1.67  | 1.95  | 2.28 | 2.66 | 3.2 |
| QC-0.3         | 5    | 4    |      | 0.3   | 0.402 | 0.538 | 0.72  | 0.964 | 1.29  | 1.73 | 2.31 | 3.1 |
| SIN < 300      | 5    | 4    | 150  | 0.35  | 0.460 | 0.604 | 0.793 | 1.04  | 1.37  | 1.80 | 2.36 | 3.1 |
| SIN 1500       | 5    | 4    | 1500 | 0.35  | 0.460 | 0.604 | 0.793 | 1.04  | 1.37  | 1.80 | 2.36 | 3.1 |

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|                |   |   |      |       |       |       |       |       |       |      |      |     |
|----------------|---|---|------|-------|-------|-------|-------|-------|-------|------|------|-----|
| TEOS < 500     | 6 | 4 | 400  | 0.35  | 0.460 | 0.604 | 0.793 | 1.04  | 1.37  | 1.80 | 2.36 | 3.1 |
| TEOS 501-1999  | 6 | 4 | 1000 | 0.35  | 0.460 | 0.604 | 0.793 | 1.04  | 1.37  | 1.80 | 2.36 | 3.1 |
| TEOS 2000-3000 | 6 | 4 | 2000 | 0.35  | 0.460 | 0.604 | 0.793 | 1.04  | 1.37  | 1.80 | 2.36 | 3.1 |
| TEOS 3001-4999 | 6 | 4 | 2000 | 0.50  | 0.628 | 0.798 | 0.991 | 1.24  | 1.56  | 1.96 | 2.47 | 3.1 |
| TEOS > 5000    | 6 | 4 | 6000 | 1     | 1.16  | 1.34  | 1.55  | 1.79  | 2.07  | 2.39 | 2.77 | 3.2 |
| WSIX-0.2UM     | 6 | 4 | 1000 | 0.232 | 0.269 | 0.358 | 0.477 | 0.636 | 0.846 | 1.13 | 1.5  | 2   |
| TIN-36         | 4 | 4 | 1000 | 0.241 | 0.47  | 0.738 | 1.16  | 1.82  | 2.85  | 4.47 | 7.01 | 11  |
| SOG-R7         | 5 | 4 | 3100 | 0.3   | 0.342 | 0.469 | 0.642 | 0.88  | 1.21  | 1.65 | 2.26 | 8   |
| WCVD-P         | 6 | 4 | 1600 | 0.175 | 0.251 | 0.329 | 0.514 | 0.737 | 1.05  | 1.51 | 2.16 | 3.1 |

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## 機台編號 Machine Number : SS-3

| PTC 程式對照表 Program Table |         |                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Recipe                  | Machine | Use                                                                                                                                                                                                                                                                                                                                                                     |
| *1                      | 6400    | BPSG in / CVD 1~6,8,9 進前/SOG 進前/WCVD 前後/EB 前後<br>BPSG in / CVD 1~6,8,9 into the top/SOG into the top/WCVD before & after/EB before & after                                                                                                                                                                                                                              |
| *5-POLY                 | 6400    | TEOS 1000A 、1500A                                                                                                                                                                                                                                                                                                                                                       |
| *7                      | 6400    | CVD 1~6 進後/SOG 進後/ Sputter PE OX 2K / SCR PE OX 2K<br>CVD 1~6 After entered/SOG after entered/ Sputter PE OX 2K/ SCR PE OX 2K                                                                                                                                                                                                                                           |
| *9                      | 6400    | SiO <sub>2</sub> > 3000A                                                                                                                                                                                                                                                                                                                                                |
| 18                      | 6400    | Pad etching                                                                                                                                                                                                                                                                                                                                                             |
| 19                      | 6400    | Metal etching/Sputter A2 E/R                                                                                                                                                                                                                                                                                                                                            |
| 20                      | 6400    | SIN for 901-3 測機 Measuring Machine                                                                                                                                                                                                                                                                                                                                      |
| *DIFF                   | 6400    | I24/I89 PTC 量測程式<br>I24/I89 PTC Measure recipe                                                                                                                                                                                                                                                                                                                          |
| *IMP                    | 6400    | RTP 測機 Measuring Machine / Mi-current 測機 Measuring Machine                                                                                                                                                                                                                                                                                                              |
| Bare-0.12μ m            | 6400    | for QC 全新 MW/Reclaim 進料檢驗<br>for QC Brand New MW/Reclaim IQC                                                                                                                                                                                                                                                                                                            |
| Bare-0.2μ m             | 6400    | QC 全新 MW/Reclaim 進料檢驗/GATE in/CV 全新控片 in/LK 全新控片 in /(RCA 、Clean 進前/進後)/Wsix 進前/GSD 測機/HFV 前後/SIN 進前 /TEOS 、HTO 進前<br>QC Brand New MW/Reclaim IQC/GATE in/CV Brand new Control Pieace in/LK Brand new control piecein/(RCA 、Clean Before/ After Entering?/Wsix before entering /GSD measuring Machine/HFV After entering/SIN before entering/ TEOS 、HTO before entering |
| BARE-SI                 | 6400    | BOE 7,9/ Oven / SCR 測機 measuring machine/E500 測機 measuring machine/封測進前 IC packaging and testing before entering                                                                                                                                                                                                                                                        |
| BPSG                    | 6400    | BPSG-5000A                                                                                                                                                                                                                                                                                                                                                              |
| CV-1000                 | 6400    | CV 1000A 全新控片 OUT<br>CV 1000A New Control Wafer OUT                                                                                                                                                                                                                                                                                                                     |
| OXIEF < 500             | 6400    | GSD 測機 Measuring Machine/ SIN Pad ox <500A                                                                                                                                                                                                                                                                                                                              |

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| SIN < 300      | 6400       | SIN < 500A                |   |       |   |
| SIN 1500       | 6400       | LP-SIN 1500A              |   |       |   |
| TEOS < 500     | 6400       | TEOS(HTO) < 500A          |   |       |   |
| TEOS 501-1999  | 6400       | LP-TEOS 500A~1999A        |   |       |   |
| TEOS 2000-3000 | 6400       | LP-TEOS 2000A/2500A/3000A |   |       |   |
| TEOS 3001-4999 | 6400       | LP-TEOS > 3000~4999A      |   |       |   |
| TEOS > 5000    | 6400       | LP-TEOS > 5000A           |   |       |   |
| POLY < 1000    | 6400       | poly < 1000A              |   |       |   |
| POLY 1001-3000 | 6400       | poly 1001A ~ 2700 A       |   |       |   |
| POLY 3001-5000 | 6400       | poly 2701A ~ 5000A        |   |       |   |
| POLY > 5001    | 6400       | poly > 5001A              |   |       |   |
| QC-0.3μ m      | 6400       | QC 進料檢驗 IQC               |   |       |   |
| TIN-36         | 6400       | Sputter 進後 after entering |   |       |   |
| SOG-R7         | 6400       |                           |   |       |   |
| WCVD-P         | 6400       |                           |   |       |   |
| WSIX-0.2UM     | 6400       | Wsix 進後 after entering    |   |       |   |

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## 機台編號Machine Number : SS-5

| PTC 程式對照表 Program Table |         |                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Recipe                  | Machine | Use                                                                                                                                                                                                                                                                                                                                                                |
| *1                      | 6420    | BPSG in / CVD 1~6,8,9 進前/SOG 進前/WCVD 前後/EB 前後<br>BPSG in / CVD 1~6,8,9 into the top/SOG into the top/WCVD before & after/EB before & after                                                                                                                                                                                                                         |
| *5-POLY                 | 6420    | TEOS 1000A 、 1500A                                                                                                                                                                                                                                                                                                                                                 |
| *7                      | 6420    | CVD 1~6 進後/SOG 進後/ Sputter PE OX 2K / SCR PE OX 2K<br>CVD 1~6 After entered/SOG after entered/ Sputter PE OX 2K/ SCR PE OX 2K                                                                                                                                                                                                                                      |
| *9                      | 6420    | SiO <sub>2</sub> > 3000A                                                                                                                                                                                                                                                                                                                                           |
| 18                      | 6420    | Pad etching                                                                                                                                                                                                                                                                                                                                                        |
| 19                      | 6420    | Metal etching/Sputter A2 E/R                                                                                                                                                                                                                                                                                                                                       |
| 20                      | 6420    | SIN for 901-3 測機 Measuring Machine                                                                                                                                                                                                                                                                                                                                 |
| *IMP                    | 6420    | RTP 測機 Measuring Machine / Mi-current 測機 Measuring Machine                                                                                                                                                                                                                                                                                                         |
| *DIFF                   | 6420    | I24/I89 PTC 量測程式<br>I24/I89 PTC Measure recipe                                                                                                                                                                                                                                                                                                                     |
| Bare-0.12μ m            | 6420    | for QC 全新 MW/Reclaim 進料檢驗<br>for QC Brand New MW/Reclaim IQC                                                                                                                                                                                                                                                                                                       |
| Bare-0.2μ m             | 6420    | QC 全新 MW/Reclaim 進料檢驗/GATE in/CV 全新控片 in/LK 全新控片 in<br>/(RCA 、 Clean 進前/進後)/Wsix 進前/GSD 測機/HFV 前後/SIN 進前<br>QC Brand New MW/Reclaim IQC/GATE in/CV Brand new Control Pieace in/LK Brand new control piecein/(RCA 、 Clean Before/ After Entering?/Wsix before entering /GSD measuring Machine/HFV After entering/SIN before entering/ TEOS 、 HTO 進前 before entering |
| BARE-SI                 | 6420    | BOE 7,9/ Oven / SCR 測機 measuring machine/E500 測機 measuring machine/封測進前 IC packaging and testing before entering                                                                                                                                                                                                                                                   |
| BPSG                    | 6420    | BPSG-5000A                                                                                                                                                                                                                                                                                                                                                         |
| CV-1000                 | 6420    | CV 1000A 全新控片 OUT<br>CV 1000A New Control Wafer OUT                                                                                                                                                                                                                                                                                                                |
| OXIEF < 500             | 6420    | GSD 測機 Measuring Machine/ SIN Pad ox <500A                                                                                                                                                                                                                                                                                                                         |

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|                |      |                           |
|----------------|------|---------------------------|
| SIN < 300      | 6400 | SIN < 500A                |
| SIN 1500       | 6420 | LP-SIN 1500A              |
| TEOS < 500     | 6420 | TEOS(HTO) <500A           |
| TEOS 501-1999  | 6420 | LP-TEOS 500A~1999A        |
| TEOS 2000-3000 | 6420 | LP-TEOS 2000A/2500A/3000A |
| TEOS 3001-4999 | 6420 | LP-TEOS >3000~4999A       |
| TEOS > 5000    | 6420 | LP-TEOS > 5000A           |
| POLY < 1000    | 6420 | poly < 1000A              |
| POLY 1001-3000 | 6420 | poly 1001A ~ 2700 A       |
| POLY 3001-5000 | 6420 | poly 2701A ~ 5000A        |
| POLY > 5001    | 6420 | poly > 5001A              |
| QC-0.3μ m      | 6420 | QC 進料檢驗 IQC               |
| TIN-36         | 6420 | Sputter 進後 after entering |
| SOG-R7         | 6420 |                           |
| WCVD-P         | 6420 |                           |
| WSIX-0.2UM     | 6420 | Wsix 進後 after entering    |

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## 伍、操作程序

### V、Operation steps

一、將晶片放置於 CASSETTE 內任 - SLOT 中，大平邊朝下，WAFER 之大平邊務必要與 CASSETTE 之下端平邊平行。

A、Place the wafer in any of the slots in the cassette with flat side facing down.  
The flat side of the wafer must be parallel to the flat side at the lower end of the cassette.

二、將欲量測之晶片放入標準晶舟（藍色 PP 材質晶舟），將此晶舟在平台上擺放妥當。

B、Place the measuring wafer in the slot of blue cassette made from PP, then place this cassette in the stage correctly.

三、如圖一在量測選擇畫面，以滑鼠選擇 RECIPE，按下滑鼠左鍵，再選擇 1 號 Load，按下滑鼠左鍵進入程式選擇畫面，如圖二。

C、Move arrowhead to the "Recipe" position and press the left key of mouse. Then move arrowhead to the "Load" position and press the left key of mouse.

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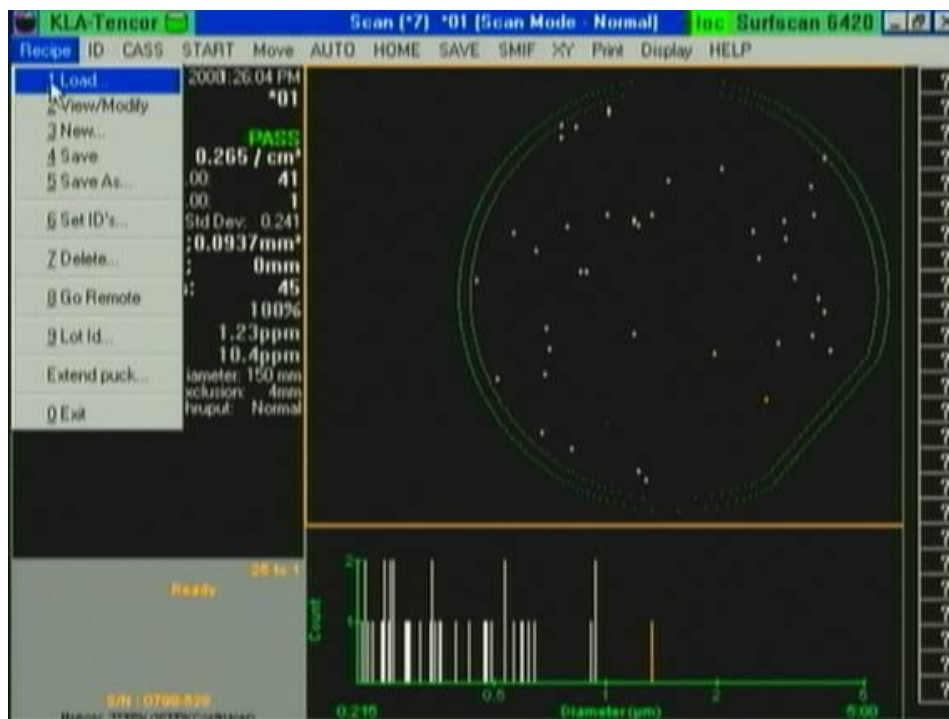


圖 (一) Pic 1

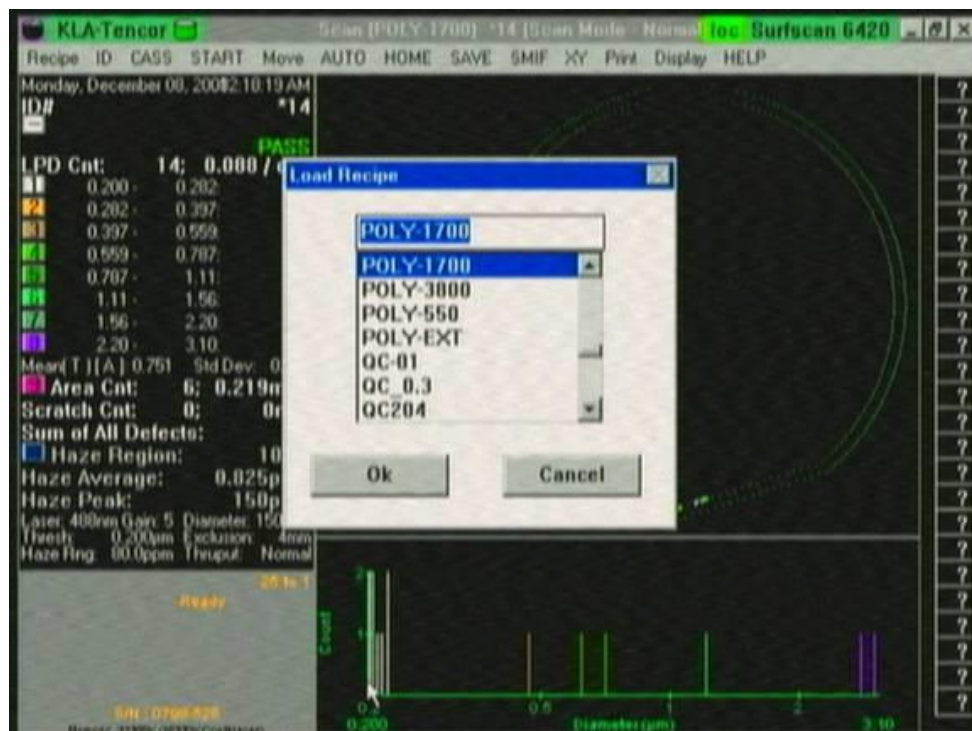


圖 (二) Pic 2

四、在圖二程式選擇欄中，以滑鼠選擇程式名稱（如 POLY-1700）按下滑鼠左鍵，當程式名稱底色

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圖案變為深藍色，再選擇畫面左下角 OK 欄位，按下滑鼠左鍵即可輸入程式，如圖三所示。

D、Move arrowhead to the measuring recipe and press the left of mouse. After the color of recipe name turn into blue, move arrowhead to the "ok" position then press the left key of mouse.

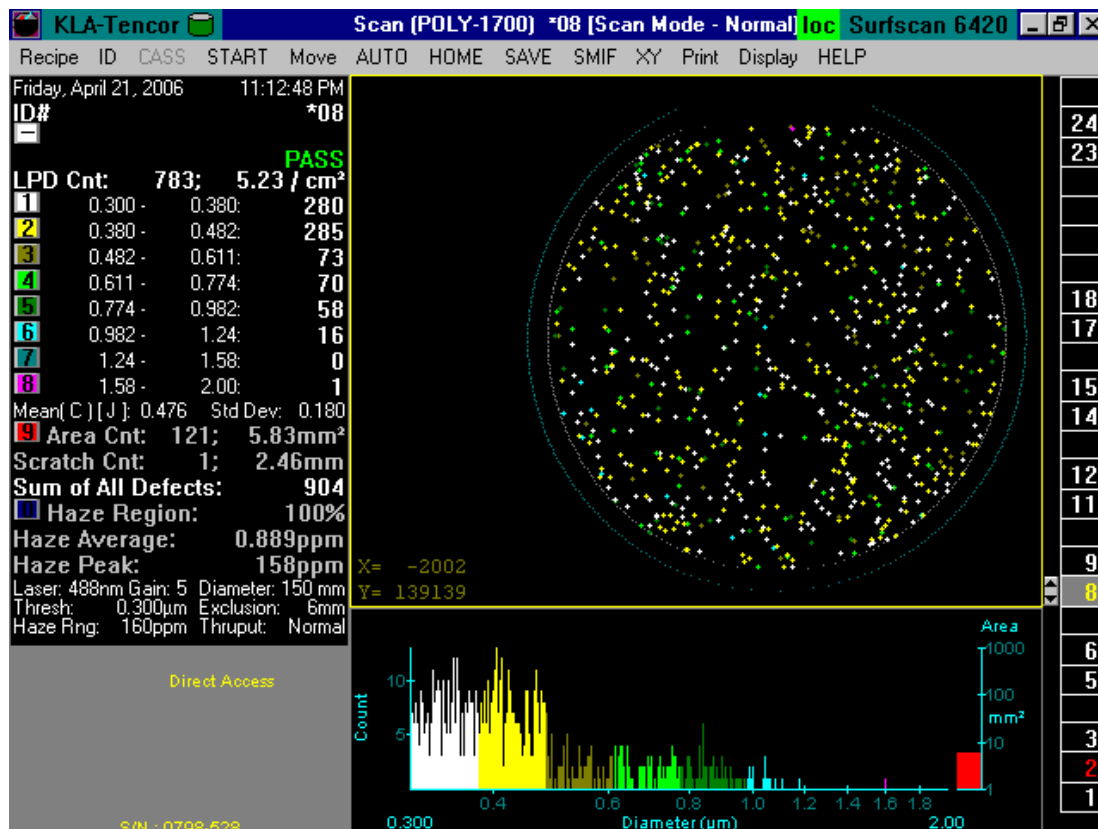


圖 ( 三 ) Pic 3

五、如圖三檢查螢幕畫面正上方出現待量測程式名稱（如 POLY-1700）字樣，即表示程式選定無誤，再以滑鼠選擇 CASS，按下滑鼠左鍵，機台會自動偵測晶舟中的晶片位置。

E、Check that the recipe name on the top of screen is the same with the recipe you Want to measure. Then move arrowhead to "CASS" position and press the left key of mouse

六、當螢幕右方出現晶舟中的晶片相對位置後，在量測晶片的位置以滑鼠按下左鍵來確認量測晶片，該位置會自動變成深灰色圖樣。

F、When the wafer position is displayed on the right side of the screen,move arrowhead

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to the wafer you want to measure and press the left key of mouse. Then it will turn into heavy-grey.

七、將滑鼠移到 START，按下滑鼠左鍵就可以進行量測。若待測晶片數量超過一片以上，應選擇 AUTO 並按下滑鼠左鍵執行多片自動量測。

G、Move arrowhead to "START" position, then press the left key of mouse and it will carry out measuring. If the number of wafer you want to measure is more than one piece, choose "AUTO" then press left key of mouse.

八、如圖四在 Sum of Dfcts 一欄中可以顯示出多片自動量測之後，各晶片 Particle 的數量，測完後，等待晶片自動送返晶舟中，以滑鼠選擇 O.K. 等待晶舟平台回到原始高度後，才可以移開晶舟。

H、You can see the particle count at "Sum of Dfcts" position after auto measuring many wafer. After wafers are loaded into cassette, press "OK" and wait for the height of cassette stage back to original height then you can remove cassette.

The screenshot shows the 'LOT SUMMARY' screen of a KLA-Tencor Surfsan 6400. The screen displays a table with columns for ID#, LPD Count, LPD / cm², Area Count, A\_Tot (mm²), Scratch Count, Scr\_Tot (mm), Sum of Dfcts, H\_Rgn (%), H\_Avg (mm), and S. The table lists data for 26 wafers. At the bottom left, 'Group Statistics' are shown. At the bottom right, a numeric keypad overlay is visible.

| ID# | LPD Count | LPD / cm² | Area Count | A_Tot (mm²) | Scratch Count | Scr_Tot (mm) | Sum of Dfcts | H_Rgn (%) | H_Avg (mm) | S  |
|-----|-----------|-----------|------------|-------------|---------------|--------------|--------------|-----------|------------|----|
| *26 | 21        | 0.100     | 16         | 2.40        | 1             | 1.76         | 37           | 66        | 0.721      | PA |
| *24 | 21        | 0.100     | 16         | 2.32        | 1             | 1.74         | 36           | 65        | 0.717      | PA |
| *20 | 44        | 0.278     | 16         | 1.80        | 0             | 0            | 59           | 65        | 0.707      | PA |
| *22 | 30        | 0.189     | 13         | 2.18        | 2             | 2.86         | 43           | 66        | 0.716      | PA |
| *21 | 29        | 0.180     | 16         | 2.12        | 0             | 0            | 44           | 65        | 0.710      | PA |
| *20 | 636       | 3.38      | 124        | 8.51        | 15            | 29.2         | 660          | 65        | 0.714      | PA |
| *19 | 37        | 0.234     | 9          | 2.09        | 1             | 2.10         | 46           | 66        | 0.720      | PA |
| *18 | 22        | 0.139     | 11         | 2.40        | 1             | 1.69         | 33           | 65        | 0.712      | PA |
| *17 | 26        | 0.168     | 14         | 2.14        | 1             | 1.11         | 39           | 65        | 0.711      | PA |
| *16 | 197       | 1.24      | 19         | 2.72        | 3             | 3.96         | 216          | 65        | 0.708      | PA |
| *16 | 96        | 0.640     | 30         | 3.80        | 3             | 5.39         | 116          | 66        | 0.729      | PA |
| *14 | 169       | 1.07      | 38         | 5.34        | 6             | 21.0         | 207          | 65        | 0.704      | PA |
| *13 | 23        | 0.146     | 14         | 2.33        | 1             | 1.44         | 37           | 65        | 0.712      | PA |
| *12 | 16        | 0.095     | 19         | 2.36        | 1             | 1.85         | 34           | 65        | 0.712      | PA |

| Group Statistics: |        |       |
|-------------------|--------|-------|
| Min:              | 15     | 0.095 |
| Max:              | 636    | 3.38  |
| Average:          | 89.64  | 0.566 |
| Std Dev:          | 140.70 | 0.888 |

|   |   |   |       |   |   |   |   |   |   |   |
|---|---|---|-------|---|---|---|---|---|---|---|
| 7 | 8 | 9 | undo  | A | B | C | D | E | F | G |
| 4 | 5 | 6 | space | H | I | J | K | L | M | N |
| 1 | 2 | 3 | clear | O | P | Q | R | S | T | U |
| 0 | . | - | enter | V | W | X | Y | Z | [ | ] |

圖 ( 四 ) Pic 4

|      |            |          |   |       |    |
|------|------------|----------|---|-------|----|
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|------|------------|----------|---|-------|----|

九、若只有一片受測晶片，如圖五在 Sum of All Defects 一欄中顯示出該晶片的 Particle 數量

。測完後，等待晶片自動送返晶舟中，以滑鼠選擇 Home，待晶舟平台回到原始高度後，才可以移開晶舟。

I、You can see the particle count at "Sun of All Dfcts" position after measuring one piece of wafer. After wafer is loaded into cassette, press "Home" and wait for height of cassette stage back to original height then you can remove cassette.

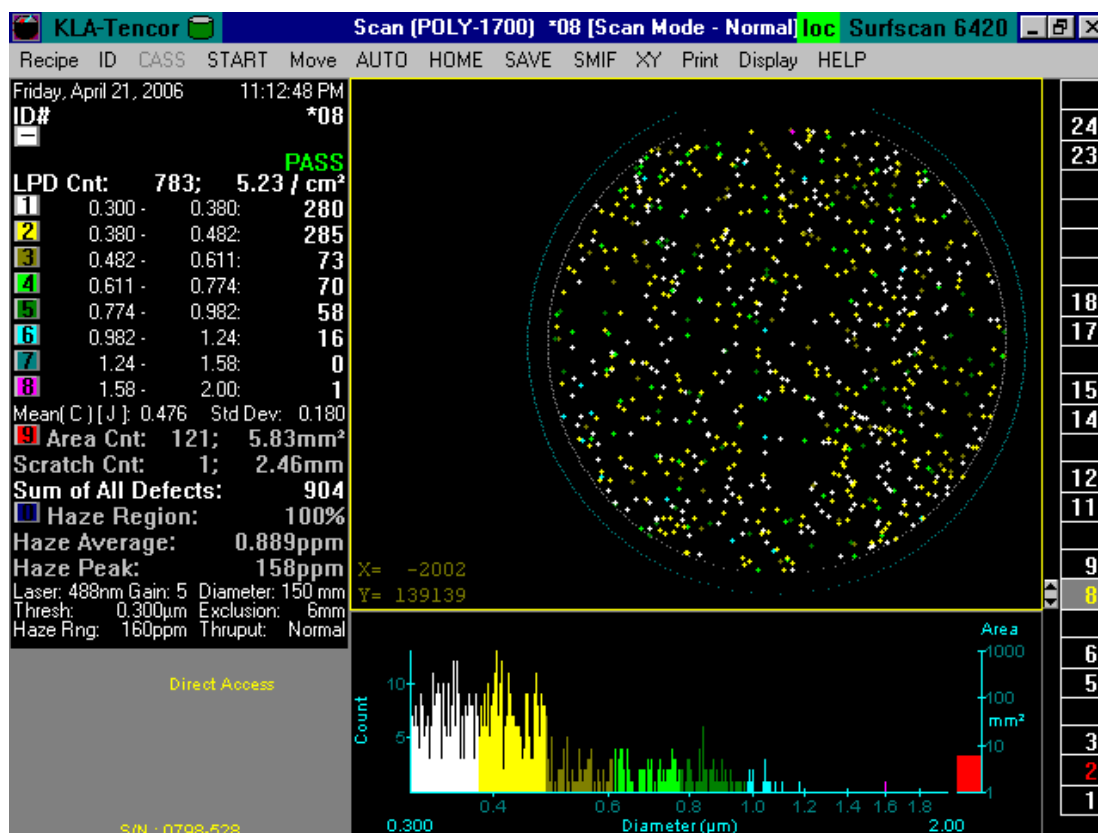


圖 ( 五 ) Pic 5

十、量測結果 Particle 超出規格時，需使用影像截取系統將圖形儲存並通知 EQ 或 PE。

影像截取操作方式見 第二十四頁至第三十一頁介紹。

J. Measurement results when a particle exceeds specifications, the system will need to use video capture and store graphic EQ notice or PE.

Video Capture mode of operation see page 24<sup>th</sup> to 31<sup>st</sup> page.

|      |            |          |   |       |    |
|------|------------|----------|---|-------|----|
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十一、擴散區控片與離子植入區控片進機台前須量測 particle，一次量一整盒，Particle 量測紀錄需使用影像截取系統將圖形儲存，存檔方式請見 第三十頁。

K. Diffusion and ion implantation zone control sheet into the machine control wafer stage area to be measured particle, the amount of time a whole box, Particle measurement records need to use video capture system graphics storage, archiving details can be found on page thirty.

|      |            |          |   |       |    |
|------|------------|----------|---|-------|----|
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|------|------------|----------|---|-------|----|

陸、自動化操作事項

## VI、automation Operation

C/P、U/T測機流程

The process for C/P,U/T

一、輸入 operator 工號及密碼

A、input work NO.and password

爐管EDC/RTC量測資料記錄/轉帳系統(AutoUT)

**nuvoTon**

Ver: 2.1.69

帳號(Account):

密碼(Password):

如有任何使用問題，08:15-17:35請電洽CIM工程師(4327/4329)或S130資管工程師(8112/8114)，其餘時間請向主任回報，謝謝 !!!\*\*\*

Exit(結束) OK(確定)

請輸入您的帳號及密碼

|      |            |          |   |       |    |
|------|------------|----------|---|-------|----|
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|------|------------|----------|---|-------|----|

二、選擇量測機台，若不存在或連線狀況不通，顯示於機台連線狀況

B、Choose the machine you want to measure.If this machine do not exist or failed to connection,automation will ask you to input the correct machine.

The screenshot shows a software window titled 'C/P --- [TEST]'. It contains three main sections: '請選擇量測機台' (Please select the machine to be measured) with a dropdown menu showing 'EM-1'; '請選擇 測機機台' (Please select the machine to be measured) with an empty dropdown menu; and '已選擇機台' (Machine already selected) with a green rectangular area. Below these is a '機台連線狀況' (Machine connection status) section with a red box containing the text '機台有誤' (Machine error). To the right of this is a button labeled '查詢測機參數' (Query machine parameters). At the bottom is a table with columns 'MWID' and 'DType'. The table has 6 rows, with the first row containing '1' and an empty cell. A warning dialog box is overlaid on the table, with a yellow warning icon and the text '無此機台連線資料, 請輸入正確之機台名稱' (No connection data for this machine, please input the correct machine name). The dialog box has a '確定' (Confirm) button. At the bottom of the window are four buttons: '放棄' (Abandon), '終止資料傳送' (Stop data transfer), '儲存資料' (Save data), and '開始轉帳' (Start accounting).

|   | MWID | DType |
|---|------|-------|
| 1 |      |       |
| 2 |      |       |
| 3 |      |       |
| 4 |      |       |
| 5 |      |       |
| 6 |      |       |

|      |            |          |   |       |    |
|------|------------|----------|---|-------|----|
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|------|------------|----------|---|-------|----|

三、選擇量測機台，連線狀況正常

C、Choose the machine you want to measure.

C/P --- [TEST]

請選擇量測機台  
EM-4

請選擇 測機機台

已選擇機台

機台連線狀況  
正常

回到功能選擇

查詢測機參數

|   | MWID | DType | Raw Data | Recipe | MEQ | wno |
|---|------|-------|----------|--------|-----|-----|
| 1 |      |       |          |        |     |     |
| 2 |      |       |          |        |     |     |
| 3 |      |       |          |        |     |     |
| 4 |      |       |          |        |     |     |
| 5 |      |       |          |        |     |     |
| 6 |      |       |          |        |     |     |

放棄 終止資料傳送 儲存資料 開始轉帳

|      |            |          |   |       |    |
|------|------------|----------|---|-------|----|
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|------|------------|----------|---|-------|----|

四、選擇測機機台，若機台狀況不允許作測機，operator 無法選擇此機台

D、Choose the process machine.If this machine status is bad,you can not choose this process machine.

C/P --- [TEST]

請選擇量測機台: EM-4

請選擇 測機機台: CVD-1A

已選擇機台:

機台連線狀況: 正常

查詢測機參數

Auto2

機台CVD-1A目前當機不能測機

確定

|   | MWID | DType | pe | MEQ | wno |
|---|------|-------|----|-----|-----|
| 1 |      |       |    |     |     |
| 2 |      |       |    |     |     |
| 3 |      |       |    |     |     |
| 4 |      |       |    |     |     |
| 5 |      |       |    |     |     |
| 6 |      |       |    |     |     |

放棄 終止資料傳送 儲存資料 開始轉帳

|      |            |          |   |       |    |
|------|------------|----------|---|-------|----|
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|------|------------|----------|---|-------|----|

五、可多選測機機台，之後按查詢測機參數，開始量測及參數對應工作

E、You can choose many process machines,then press "inquire the parameter for test run".It will start to measure.

C/P --- [TEST]

請選擇量測機台: EM-4

請選擇 測機機台: CVD-2B

已選擇機台: CVD-2A, CVD-2B

機台連線狀況: 正常

回到功能選擇

查詢測機參數

|   | MWID | DType | Raw Data | Recipe | MEQ | wno |  |
|---|------|-------|----------|--------|-----|-----|--|
| 1 |      |       |          |        |     |     |  |
| 2 |      |       |          |        |     |     |  |
| 3 |      |       |          |        |     |     |  |
| 4 |      |       |          |        |     |     |  |
| 5 |      |       |          |        |     |     |  |
| 6 |      |       |          |        |     |     |  |

放棄 終止資料傳送 儲存資料 開始轉帳

|      |            |          |   |       |    |
|------|------------|----------|---|-------|----|
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|------|------------|----------|---|-------|----|

六、Operator 開始量測

F、Operator start to measure.

The screenshot shows the CVD-2A software interface. At the top, there are tabs for CVD-2A and CVD-2B. Below the tabs is a table with columns: parameter name, 參數名稱, 參數值, and 原始資料. The table contains 7 rows of parameters. Below the table is a section with columns: MWID, DType, Raw Data, Recipe, MEQ, and wno. The Raw Data column is currently displaying a blue message: 等待儀器量測資料中, 請稍候. At the bottom of the interface are four buttons: 放棄, 終止資料傳送, 儲存資料, and 開始轉帳.

|   | parameter name | 參數名稱      | 參數值 | 原始資料 |
|---|----------------|-----------|-----|------|
| 1 | PARTICLE-      | PARTICLE  |     | 前後   |
| 2 |                |           |     |      |
| 3 | OX-E-R         | OX-E-R    |     |      |
| 4 | OX-STRESS      | OX-STRESS |     |      |
| 5 | OX-THK         | OX-THK    |     |      |
| 6 | OX-THK-U       | OX-THK-U  |     |      |
| 7 | OX-RI          | OX-RI     |     |      |

|   | MWID | DType | Raw Data       | Recipe | MEQ | wno |
|---|------|-------|----------------|--------|-----|-----|
| 1 |      |       | 等待儀器量測資料中, 請稍候 |        |     |     |
| 2 |      |       |                |        |     |     |
| 3 |      |       |                |        |     |     |
| 4 |      |       |                |        |     |     |
| 5 |      |       |                |        |     |     |
| 6 |      |       |                |        |     |     |

放棄      終止資料傳送      儲存資料      開始轉帳

|      |            |          |   |       |    |
|------|------------|----------|---|-------|----|
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|------|------------|----------|---|-------|----|

七、量出之 raw data 會即時顯示於下方表格內

G、The raw data will be displayed in the form below.

The screenshot shows a software window titled "C/P --- [TEST]". It contains two main sections. The top section is a table with columns: "parameter name", "參數名稱", "參數值", and "原始資料". The bottom section is a table with columns: "MWID", "DType", "Raw Data", "Recipe", "MEQ", "wno", and a final column with a scroll bar. Below the tables are four buttons: "放棄", "終止資料傳送", "儲存資料", and "開始轉帳".

|   | parameter name | 參數名稱      | 參數值 | 原始資料 |
|---|----------------|-----------|-----|------|
| 1 | PARTICLE-      | PARTICLE  |     | 前後   |
| 2 |                |           |     |      |
| 3 | OX-E-R         | OX-E-R    |     |      |
| 4 | OX-STRESS      | OX-STRESS |     |      |
| 5 | OX-THK         | OX-THK    |     |      |
| 6 | OX-THK-U       | OX-THK-U  |     |      |
| 7 | OX-RI          | OX-RI     |     |      |

|   | MWID | DType | Raw Data                       | Recipe | MEQ  | wno |   |
|---|------|-------|--------------------------------|--------|------|-----|---|
| 1 |      | R     | 5196.17 4790.87 4894.48 4991.1 |        | EM-5 | 1   | 5 |
| 2 |      | R     | 5018.93 4928.89 4989.68 5193.5 |        | EM-5 | 2   | 5 |
| 3 |      | R     | 5114.41 4932.92 4831.82 4787.4 |        | EM-5 | 3   | 5 |

放棄      終止資料傳送      儲存資料      開始轉帳

|      |            |          |   |       |    |
|------|------------|----------|---|-------|----|
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|------|------------|----------|---|-------|----|

八、Operator 將下方量測資料對應至某參數列，系統自動計算參數值

H、Press the left key of mouse at the parameter form and parameter data will be caculated automatically by automation system.

The screenshot shows a software window titled "C/P --- [TEST]". It contains two main data tables and a set of control buttons at the bottom.

**Table 1: Parameter Data**

|   | parameter name | 參數名稱      | 參數值      | 原始資料               |
|---|----------------|-----------|----------|--------------------|
| 1 | PARTICLE-      | PARTICLE  |          | 前後                 |
| 2 |                |           |          |                    |
| 3 | OX-E-R         | OX-E-R    |          |                    |
| 4 | OX-STRESS      | OX-STRESS |          |                    |
| 5 | OX-THK         | OX-THK    | 4965.878 | 5196.17 4790.87 48 |
| 6 | OX-THK-U       | OX-THK-U  |          |                    |
| 7 | OX-RI          | OX-RI     |          |                    |

**Table 2: Raw Data**

|   | MWID | DType | Raw Data                       | Recipe | MEQ | wno |
|---|------|-------|--------------------------------|--------|-----|-----|
| 1 |      | R     | 5196.17 4790.87 4894.48 4991.1 | EM-5   | 1   | 5   |
| 2 |      | R     | 5018.93 4928.89 4989.68 5193.5 | EM-5   | 2   | 5   |
| 3 |      | R     | 5114.41 4932.92 4831.82 4787.4 | EM-5   | 3   | 5   |

**Buttons:** 放棄 (Abandon), 終止資料傳送 (Stop Data Transfer), 儲存資料 (Save Data), 開始轉帳 (Start Accounting)

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九、對應另一測機機台同類參數

I、Link the same type parameter of the other process machine.

C/P --- [TEST]

CVD-2A CVD-2B

|   | parameter name | 參數名稱      | 參數值      | 原始資料               |
|---|----------------|-----------|----------|--------------------|
| 1 | PARTICLE-      | PARTICLE  |          | 前後                 |
| 2 |                |           |          |                    |
| 3 | OX-E-R         | OX-E-R    |          |                    |
| 4 | OX-STRESS      | OX-STRESS |          |                    |
| 5 | OX-THK         | OX-THK    | 5028.012 | 5018.93 4928.89 49 |
| 6 | OX-THK-U       | OX-THK-U  |          |                    |
| 7 | OX-RI          | OX-RI     |          |                    |

|   | MWID | DType | Raw Data                       | Recipe | MEQ | wno |
|---|------|-------|--------------------------------|--------|-----|-----|
| 1 |      | R     | 5196.17 4790.87 4894.48 4991.1 | EM-5   | 1   | 5   |
| 2 |      | R     | 5018.93 4928.89 4989.68 5193.5 | EM-5   | 2   | 5   |
| 3 |      | R     | 5114.41 4932.92 4831.82 4787.4 | EM-5   | 3   | 5   |

放棄 終止資料傳送 儲存資料 開始轉帳

|      |            |          |   |       |    |
|------|------------|----------|---|-------|----|
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|------|------------|----------|---|-------|----|

十、Operator 先儲存已對應好的參數值

J、Save the parameter data.

十一、Operator 開始另一種類之參數量測流程

K、Operator start to measure the other parameter.

請選擇量測機台: SS-1

請選擇 測機機台: CVD-2B

已選擇機台: CVD-2A, CVD-2B

機台連線狀況: 正常

回到功能選擇

查詢測機參數

|   | MWID | DType | Raw Data | Recipe | MEQ | wno |
|---|------|-------|----------|--------|-----|-----|
| 1 |      |       |          |        |     |     |
| 2 |      |       |          |        |     |     |
| 3 |      |       |          |        |     |     |
| 4 |      |       |          |        |     |     |
| 5 |      |       |          |        |     |     |
| 6 |      |       |          |        |     |     |

放棄 終止資料傳送 儲存資料 開始轉帳

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十二、之前存入的參數值會自動帶出

L、The parameter data saved before will be displayed automatically.

C/P --- [TEST]

**CVD-2A** CVD-2B

|   | parameter name | 參數名稱      | 參數值      | 原始資料           |
|---|----------------|-----------|----------|----------------|
| 1 | PARTICLE-      | PARTICLE  |          | 前後<br>4965.878 |
| 2 |                |           |          |                |
| 3 | OX-E-R         | OX-E-R    |          |                |
| 4 | OX-STRESS      | OX-STRESS |          |                |
| 5 | OX-THK         | OX-THK    | 4965.878 |                |
| 6 | OX-THK-U       | OX-THK-U  |          |                |
| 7 | OX-RI          | OX-RI     |          |                |

|   | MWID | DType | Raw Data      | Recipe | MEQ | wno |
|---|------|-------|---------------|--------|-----|-----|
| 1 |      |       | 等待機臺量測資料中,請稍候 |        |     |     |
| 2 |      |       |               |        |     |     |
| 3 |      |       |               |        |     |     |
| 4 |      |       |               |        |     |     |
| 5 |      |       |               |        |     |     |
| 6 |      |       |               |        |     |     |

放棄 終止資料傳送 儲存資料 開始轉帳

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十三、進行其他參數值對應

M、Start to link the other parameter.

CP --- [TEST]

**CVD-2A** CVD-2B

|   | parameter name | 參數名稱      | 參數值      | 原始資料     |
|---|----------------|-----------|----------|----------|
| 1 | PARTICLE-      | PARTICLE  | 1        | 前 0      |
| 2 |                |           |          | 後 1      |
| 3 | OX-E-R         | OX-E-R    |          |          |
| 4 | OX-STRESS      | OX-STRESS |          |          |
| 5 | OX-THK         | OX-THK    | 4965.878 | 4965.878 |
| 6 | OX-THK-U       | OX-THK-U  |          |          |
| 7 | OX-RI          | OX-RI     |          |          |

|   | MWID | DType | Raw Data | Recipe | MEQ  | wno |   |
|---|------|-------|----------|--------|------|-----|---|
| 1 |      | P     | 0        |        | SS-3 | 1   | 1 |
| 2 |      | P     | 4        |        | SS-3 | 2   | 1 |
| 3 |      | P     | 1        |        | SS-3 | 3   | 1 |
| 4 |      | P     | 2        |        | SS-3 | 4   | 1 |

放棄 終止資料傳送 儲存資料 開始轉帳

|      |            |          |   |       |    |
|------|------------|----------|---|-------|----|
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|------|------------|----------|---|-------|----|

十四、所有此次測機所須的參數都對應完成後開始進行轉帳

N、You must link all of the parameters for this testing run,then you can start to move out.

C/P --- [TEST]

CVD-2A

CVD-2B

|   | parameter name | 參數名稱      | 參數值      | 原始資料     |  |
|---|----------------|-----------|----------|----------|--|
| 1 | PARTICLE-      | PARTICLE  | 1        | 前 0      |  |
| 2 |                |           |          | 後 1      |  |
| 3 | OX-E-R         | OX-E-R    |          |          |  |
| 4 | OX-STRESS      | OX-STRESS |          |          |  |
| 5 | OX-THK         | OX-THK    | 4965.878 | 4965.878 |  |
| 6 | OX-THK-U       | OX-THK-U  |          |          |  |
| 7 | OX-RI          | OX-RI     |          |          |  |

|   | MWID | DType | Raw Data | Recipe | MEQ  | wno |  |
|---|------|-------|----------|--------|------|-----|--|
| 1 |      | P     | 0        |        | SS-3 | 1 1 |  |
| 2 |      | P     | 4        |        | SS-3 | 2 1 |  |
| 3 |      | P     | 1        |        | SS-3 | 3 1 |  |
| 4 |      | P     | 2        |        | SS-3 | 4 1 |  |

放棄

終止資料傳送

儲存資料

開始轉帳

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十五、可進行所有機台的測機參數收集，或是先進行部份機台的收集

O、You can collect the parameters of testing run for all machine. Also you can collect the parameters of testing run for part of machine.

|   | parameter name | 參數名稱      | 參數值      | 原始資料     |
|---|----------------|-----------|----------|----------|
| 1 | PARTICLE-      | PARTICLE  | -2       | 前 4      |
| 2 |                |           |          | 後 2      |
| 3 | OX-E-R         | OX-E-R    |          |          |
| 4 | OX-STRESS      | OX-STRESS |          |          |
| 5 | OX-THK         | OX-THK    | 5028.012 | 5028.012 |
| 6 | OX-THK-U       | OX-THK-U  |          |          |
| 7 | OX-RI          | OX-RI     |          |          |

☒ CVD-2A  
☒ CVD-2B

開始進行轉帳

回到參數對應

離開測機流程

## Sheet 測機流程

The process for sheet testing run

一、輸入 operator 工號及密碼

A、Input work NO. and password.

二、輸入 sheet id 和控片代號

B、Input sheet ID and monitor wafer code.

三、可正向或逆向連續量測

C、You can measure wafer successively in order or not in orders.

四、可只量部分控片

D、You can just measure part of monitor wafers.

The screenshot shows the 'Virtual Sheet -- [s100001]' window. It has two tabs: '輸入控片代碼' (Input Wafer Code) and '確定&儲存' (Confirm & Save). The '輸入控片代碼' tab is active. It contains the following fields and controls:

- 量測機台** (Measurement Machine): A dropdown menu showing 'SS-1'.
- 機台連線狀況** (Machine Connection Status): A button labeled '正常' (Normal).
- SHEET ID**: A text box containing 'mysheet'.
- 控片代碼** (Wafer Code): A text box containing '4-147'.
- 測量模式** (Measurement Mode): Two radio buttons. '連續量測' (Continuous Measurement) is selected, and '各別量測' (Individual Measurement) is unselected.
- 從第 25 片到第 1 片** (From Wafer 25 to Wafer 1): A range selection field with '25' in the '從第' box and '1' in the '到第' box.
- Wafer Grid**: A 5x5 grid of buttons numbered 1 to 25. Buttons 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, and 25 are all highlighted in red.
- Buttons**: Three buttons at the bottom: '重新輸入' (Re-enter), '結束' (End), and '確定' (Confirm).

五、儲存顯示之量測值

E、Save the measurement data

Virtual Sheet -- [s100001]

輸入控片代碼 確定&儲存

原有資料

量測資料

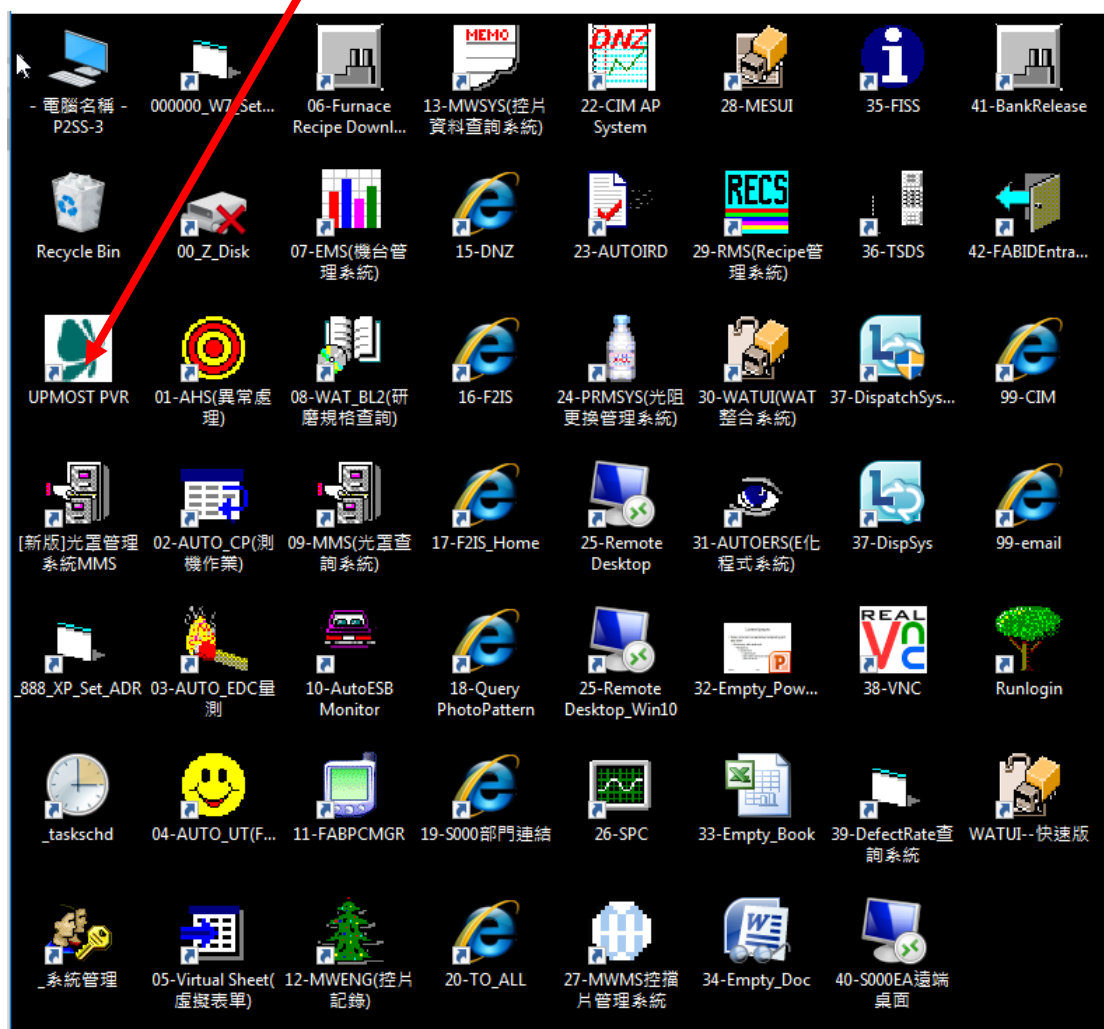
|                | MWID     | Dtype | Raw Data | Recipe | MEQ |
|----------------|----------|-------|----------|--------|-----|
| 等待機臺量測資料中, 請稍候 |          |       |          |        |     |
| 1              | 4-147#25 |       |          |        |     |
| 2              | 4-147#24 |       |          |        |     |
| 3              | 4-147#23 |       |          |        |     |
| 4              | 4-147#22 |       |          |        |     |
| 5              | 4-147#21 |       |          |        |     |
| 6              | 4-147#20 |       |          |        |     |
| 7              | 4-147#19 |       |          |        |     |
| 8              | 4-147#18 |       |          |        |     |

結束 Sheet 流程 回到前一畫面 停止資料傳送 儲存 Sheet

|      |            |          |   |       |    |
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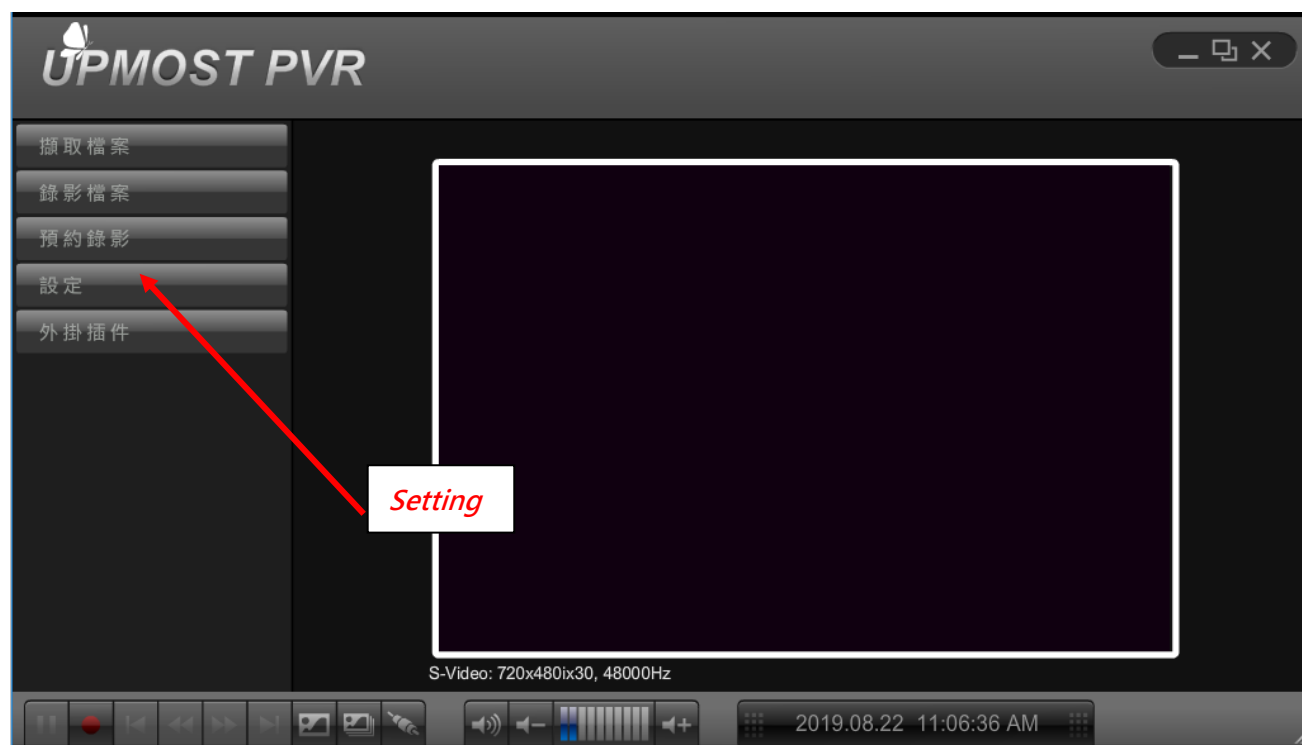
## 影像截取操作方式 *Video Capture Operation*

### 一、點選 *Click UPMOST PVR.*



|      |            |          |   |       |    |
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## 二、點選設定 Click Set Up



## 三、點選截圖設定 Click Screenshot setting



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## 四、截圖檔案命名規則如下 Screenshot file name rule as following item.

1. Particle UP 請輸入“機台、日期、時間、位置”等資訊。例：LP-P6 11月6日 9點 30分

C1 Particle UP (輸入方式 (20190905 LP-P6 C1))，按確定。

Particle UP please enter “machine, date, time and location”, and so on, Example:

LP-P6 September 6, at 9:30 C1 Particle UP (input mode (20190905 LP-P6 C1)), press ok.

2. Particle 控片及 LK 控片請輸入“控片編號、SLOT 位置”等資訊。例：1-006 #1~#25(輸入方式 (1-006\_1-25))。

Particle control and LK control please enter “control number, SLOT locations”, and so on, Example: 1-006#1~#25(Input mode (1-006\_1-25)).

3. CV 控片請輸入“控片刻號、SLOT 位置”等資訊。例：CV-XX #1~#25 (輸入方式 (CV-XX\_1-25))。

註：CV-XX (XX 為 01~99)

CV control, SLOT locations, please enter “control for a moment”, and so on,

Example: CV-XX #1~#25 (Input mode (CV-XX\_1-25), Note: the CV-XX (XX-01-99)

4. 離子植入區控片請輸入“控片刻號、SLOT 位置”等資訊。例：U-IMP-N-XX #1~#25 或

U-IMP-P-XX #1~#25 (輸入方式 (IMP-N-XX\_1-25) 或 (IMP-P-XX\_1-25))。

註：U-IMP-N-XX (XX 為 01~99)

Ion-implanted region control, SLOT locations, please enter “control for a moment”, and so on, Example: U-IMP-N-XX or #1~#25 U-IMP-P-XX #1~#25 (input mode (IMP-N-XX\_1-25) or (IMP-P-XX\_1-25)).

Note: the U-IMP-N-XX (XX-01~99))

|      |            |          |   |       |    |
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## 五、截圖檔案命名設定方式 Screenshot file name setting method.

### 1. 儲存資料夾嚴禁修改

#### 1.Do not change folder path.

### 2. 截圖檔案命名請輸入相關資訊。如 “機台、日期、時間、位置” 等

#### 2. please enter “machine, date, time and location” , and so on.

例：LP-P6 09月05日 9點 30分 C1 Particle UP (輸入方式 (20190905 LP-P6 C1))，按確定。

Example: LP-P6 September 5, at 9:30 C1 Particle UP (input mode (20190905 LP-P6 C1))

### 3.修改完成按確認 Press ok after modified.

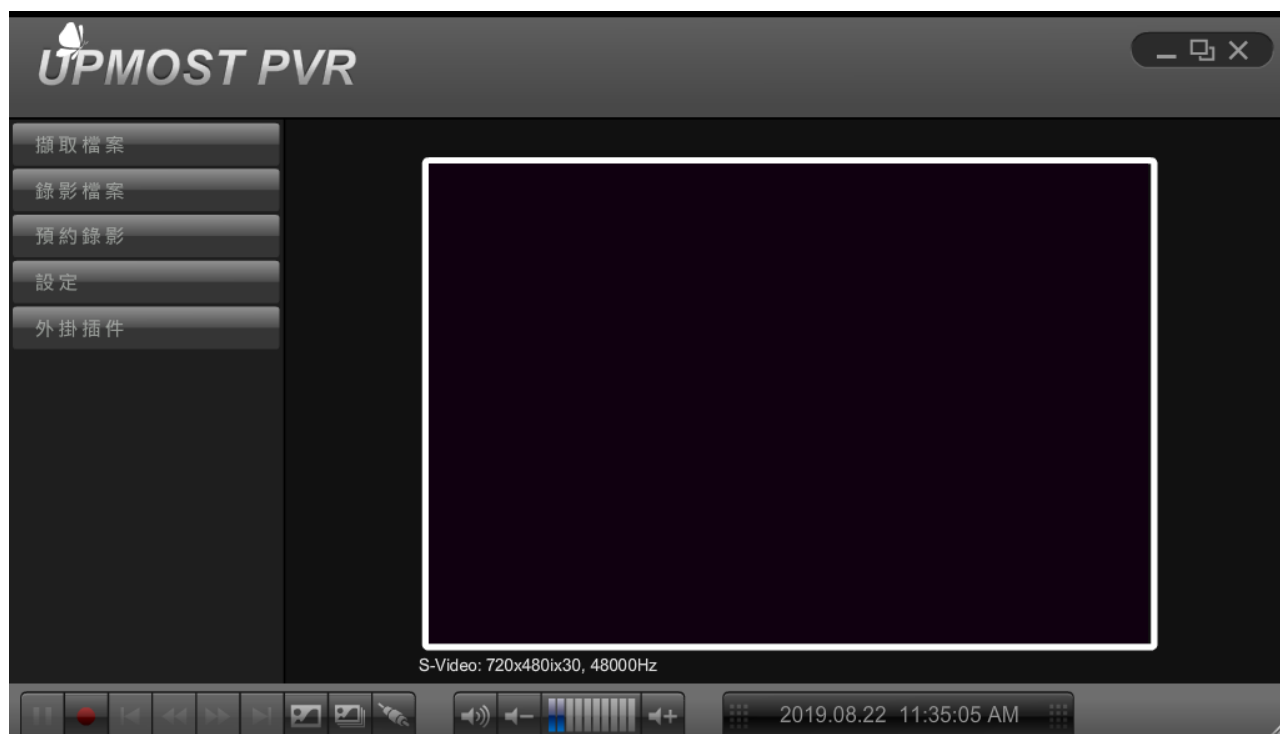


|      |            |          |   |       |    |
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### 3. 點選返回 Click return.

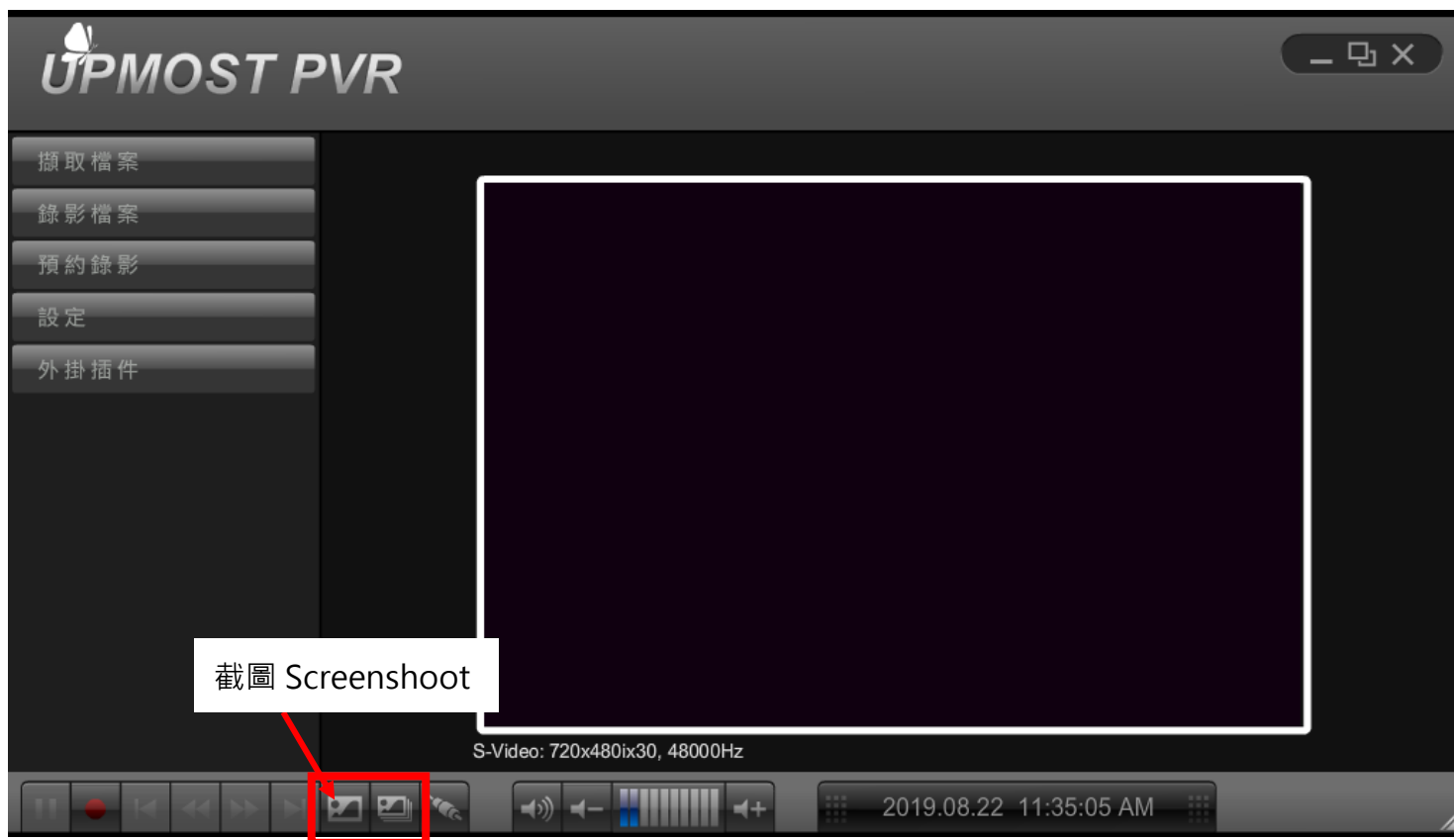


### 4. 截圖檔案命名完成 Screenshot file name already set up.



|      |            |          |   |       |    |
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六、掃描晶片完成後，點選截圖 After scanning the wafer is completed, click Screenshot



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## 柒、操作注意事項

### VII、Cautions

一、嚴禁震動機台並禁止堆放雜物在機台上。

A、NO shaking machine and don't lay others at machine.

二、機台下方門板禁止開啟，以避免誤觸電源等開關。

B、Don't open the door at the bottom of machine and it will keep you from getting an electric shock.

三、嚴禁離開至本 O.I. 所規定以外之畫面，如因軟體問題而跳離，請勿任意動作，並儘速洽工程師處理。

C、It is banned to go to the screen which is not permitted by O.I. If you find the screen which is not permitted by O.I., do not take action and contact PE immediately.

四、嚴禁擅自關閉及開啟電源，以免造成軟、硬體上之損壞。

D、Don't turn on and turn off power supply and it will keep software and hardware from

五、嚴禁擅自更改程式內容，造成量測誤差(如下附圖)，如發現(附圖二)顏色反灰洽工程師處理。

E、Forbid ro change the content of recipes, if you find the image' s color is gray, please call PE to check.(see the attachment figure)

NO.:

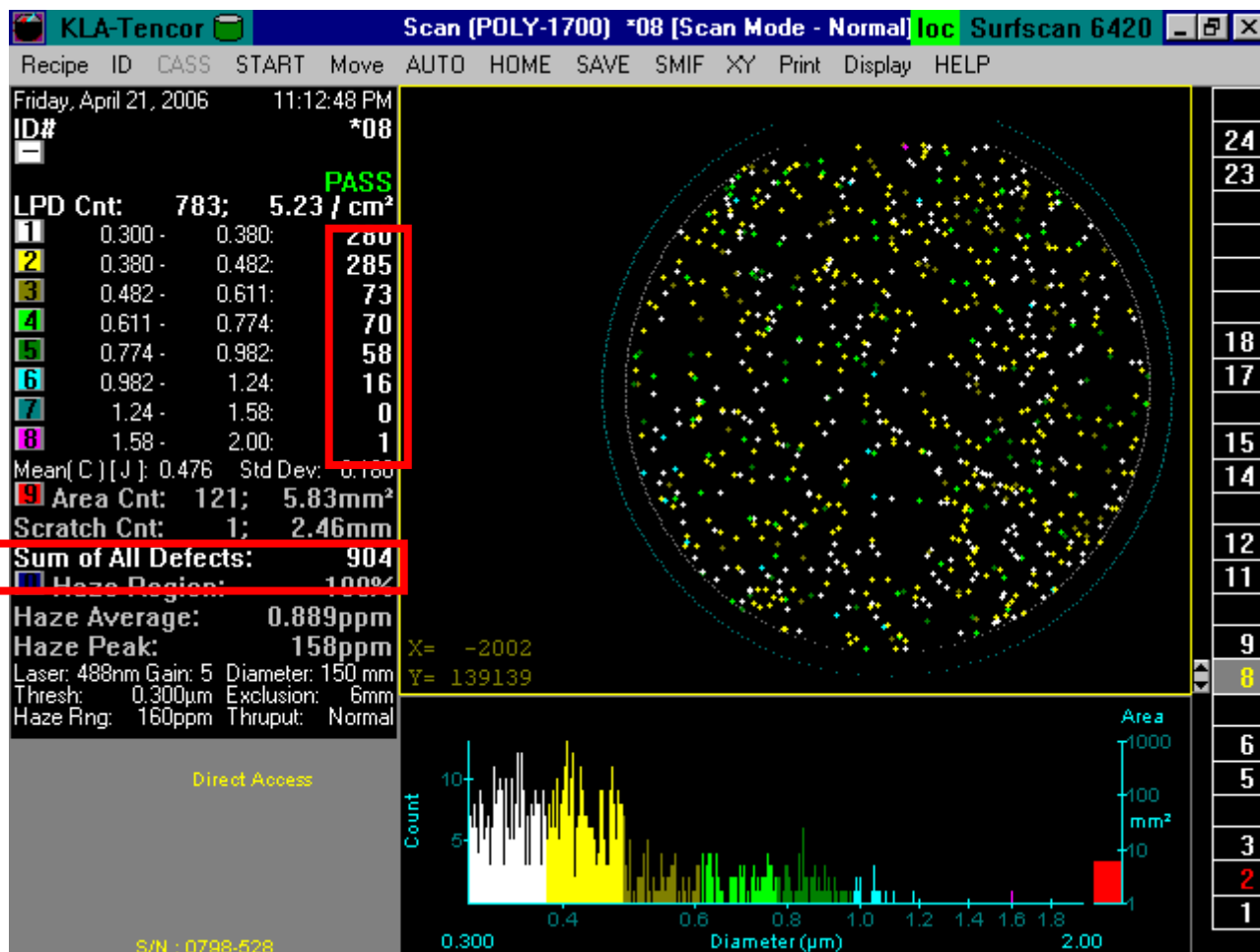
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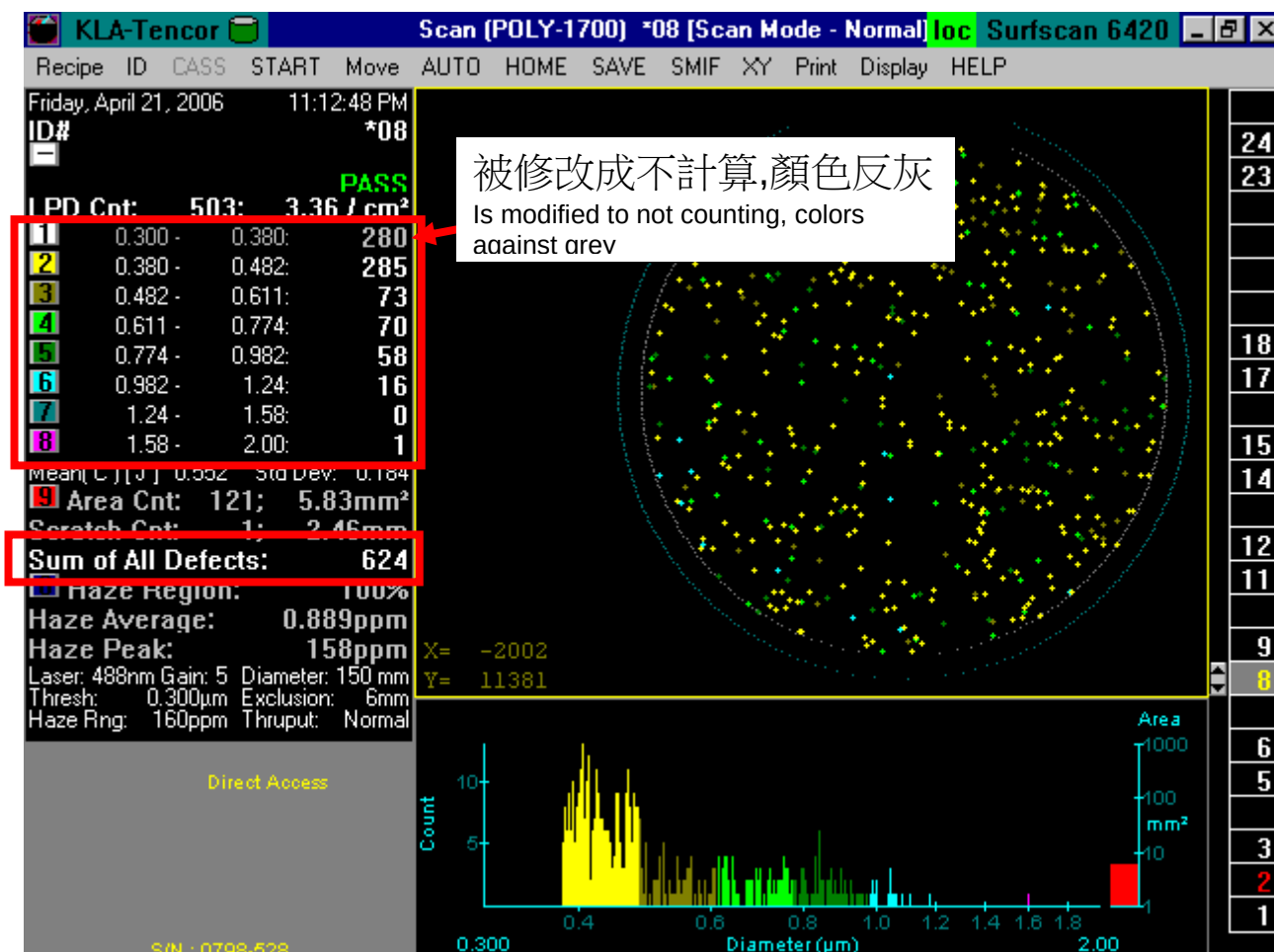
S

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(附圖一)Reference 1



(附圖二)Reference 2

六、晶片卡在機台內應通知工程師處理。

F、If wafer is caught in the machine, please contact EQ.

七、只有在綠燈亮時才可移動晶舟，綠燈未亮表示晶舟未回到 HOME 傾斜位置，此時嚴禁移動晶舟，以免造成手臂折斷。

G、When green light is turned on, you just can move cassette. When green light is not turned on, it means cassette is not yet back to the "Home" postition and it is banned to move cassette, otherwise it will break the robot.

|      |            |          |   |       |    |
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## 捌、品質控制

### VIII、Quality Control

一、本機台每個月由製程人員以 VLSI 標準片進行校驗量測，如不合規格須停機進行調整。

A、Process engineer must measure VLSI standard wafers to calculate the machine every month. If the data are out-of-spec, PE should adjust this machine instantly.

二、儀器定期校驗：(Periodic calibration)

1.校驗執行人：PE

Performed by: PE

校驗頻率：6個月

Frequency: 6 months

標準件：VLSI 提供之標準片 ACS6-0.204。

Standard: VLSI provided standard wafer ACS6-0.204.

2.校驗報告傳至儀校中心。

The original of calibration report will be forwarded to Instrument Calibration Center.

3.校驗結果不合規格時,由製程工程師通知設備工程師或代理商人員調整。

重新校驗結果合規格後,機台始能繼續上線量測。

If the calibration result is found not up to specs, process engineering will notify EQ engineer or vendor for adjustment.

二、若每次量測，在固定位置會出現特定形狀之 PARTICLE（色點），此時速通知 PE CHECK。

B、If you find the same shape of particle in the same area when you measure, please contact PE immediately.

三、定期儀器之校正時機依儀器經歷卡上之校正週期執行校正，量測標準片比對量測值與標準值並記錄於校正報告上，依經歷卡上的容許誤差值進行是否校正合格之判定，並依

“0000-1000 儀器校驗管理程序”執行相關校驗規定。

C. The timing of periodic instrument calibration correction is performed according to the calibration of

|      |            |          |   |       |    |
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the instrument through the card of the cycle, measurement standard sheet than the measured value and standard

The reference value and recorded on the calibration report, according to the allowable error value through the card will be determined whether the corrected qualified, and by

"0000-1000 instrument calibration management program" implementation of the relevant provisions of the checksum

## 玖、生效與修訂

### IX. Effective and Revision

本規範之公佈實施及其修訂核准層級皆依會簽/核決/分發依循範例為之。

Its promulgation and implementation of the revised specification levels are approved Countersigned by / nuclear decision / Distribution according to Follow Examples whom.